

Homework 10

Grant Talbert

November 14, 2025

Boston University

Remark 0.1. Since all problems are optional, I had fun with the solutions where possible. I was also more hand-wavy with proofs.

Problem 1

Problem 1.1. Let M be an R -module. Define the annihilator

$$\text{Ann}_R(M) := \{r \in R \mid \forall m \in M, rm = 0\}.$$

Show $\text{Ann}_R(M)$ is an ideal of R .

Proof. An R -module can be defined as an abelian group M and a ring homomorphism $\rho : R \rightarrow \text{End}_{\text{Ab}}(M)$, where we define $rm := \rho(r)(m)$. The annihilator $\text{Ann}_R(M)$ is then by definition the kernel of ρ , and since ρ is a ring homomorphism, its kernel is an ideal. ■

Problem 1.2. Find a module with a trivial annihilator.

Proof. The $\mathbb{R}$ -vector space $\mathbb{R}$ has trivial annihilator because $xy = 0$ if and only if one of x or y is 0. More generally, $\text{Ann}_D(D)$ for any integral domain D is trivial. ■

Problem 1.3. Find a module with nontrivial annihilator.

Proof. $\mathbb{Z}/2\mathbb{Z}$ as a $\mathbb{Z}$ -module has nontrivial annihilator. Consider $2 \in \mathbb{Z}$. We have $2 \cdot 0 = 0$ and $2 \cdot 1 = 1 + 1 = 0$, so $2 \in \text{Ann}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z})$. ■

Problem 2

Problem 2.1. Let R a ring and M, N R -modules. Show the following are R -modules.

1. $R^{\oplus A}$; A a set;
2. $M_{m \times n}(R)$;
3. $R\text{-Mod}(M, N)$.

Determine which ones are free.

Proof. (1) Done in class.

(2) Since we are dealing with matrix addition, there is an isomorphism of abelian groups $\varphi : R^{mn} \cong R^{\oplus mn} \rightarrow M_{m \times n}(R)$ for any choice of basis in $R^{\oplus mn}$ by choosing an ordering for the matrix components, and inserting the i th entry into the i th slot of the matrix. Simply define the R -module structure on $M_{m \times n}(R)$ such that this isomorphism promotes to one of R -modules; meaning for any $X \in M_{m \times n}(R)$ and $r \in R$,

$$rX := \varphi(r\varphi^{-1}(X)).$$

This clearly agrees with the usual pointwise definition. Moreover, the module is free as an R -module.

(3) This only holds if R is commutative. Let $\varphi : M \rightarrow R$, $m \in M$, and $r, s \in R$. For $r\varphi$ to be R -linear, we must have $(r\varphi)(sm) = s(r\varphi)(m)$. However, since φ is linear, we have $(r\varphi)(sm) = r(\varphi(sm)) = r(s\varphi(m)) = (rs)\varphi(m)$. We thus require that $rs = sr$.

Define a ring R as a monoid object in $\mathcal{A}b$. An R -module is then an abelian group homomorphism $\rho : R \otimes_{\mathbb{Z}} M \rightarrow M$; this agrees with the usual definition since we require that the map $\rho : R \times M \rightarrow M$ be $\mathbb{Z}$ -bilinear, and thus factors uniquely as a $\mathbb{Z}$ -linear map via the tensor product. We define an R -module structure on $R\text{-Mod}(M, N)$ as follows. Define λ as the composite

$$\lambda : R \otimes_{\mathbb{Z}} \mathcal{A}b(M, N) \otimes_{\mathbb{Z}} M \xrightarrow{1_R \otimes \text{ev}} R \otimes_{\mathbb{Z}} N \xrightarrow{\rho_N} N,$$

where $\text{ev} : \mathcal{A}b(M, N) \otimes_{\mathbb{Z}} M \rightarrow N$ is the evaluation map $(\varphi, m) \mapsto \varphi(m)$, which is $\mathbb{Z}$ -bilinear as a map $\mathcal{A}b \times M \rightarrow N$, and thus is well-defined as a $\mathbb{Z}$ -linear map from the tensor product. Define $\mu : R \otimes_{\mathbb{Z}} \mathcal{A}b(M, N) \rightarrow \mathcal{A}b(M, N)$ as the transpose of λ under $- \otimes_{\mathbb{Z}} M \dashv \mathcal{A}b(M, -)$. We claim μ induces an R -module structure on $R\text{-Mod}(M, N)$ by restricting along $R\text{-Mod}(M, N) \subseteq \mathcal{A}b(M, N)$.

To check, this, let η and ε be the unit and counit, respectively, of the adjunction $- \otimes_{\mathbb{Z}} M \dashv \mathcal{A}b(M, -)$. There are two useful identities this provides:

$$\mu = \mathcal{A}b(M, \lambda) \circ \eta_{R \otimes_{\mathbb{Z}} \mathcal{A}b(M, N) \otimes_{\mathbb{Z}} M},$$

$$\lambda = \varepsilon_{\mathcal{A}b(M, N)} \circ (\mu \otimes_{\mathbb{Z}} 1_M).$$

a

Writing $m_R : R \times R \rightarrow R$ for ring multiplication and α for the associator for tensor, associativity is expressed as the commutativity of the pentagon diagram

$$\begin{array}{ccc} (R \otimes_{\mathbb{Z}} R) \otimes_{\mathbb{Z}} \mathcal{A}b(M, N) & \xrightarrow{m_R \otimes 1} & R \otimes_{\mathbb{Z}} \mathcal{A}b(M, N) \\ \alpha \downarrow & & \downarrow \mu \\ R \otimes_{\mathbb{Z}} (R \otimes_{\mathbb{Z}} \mathcal{A}b(M, N)) & \xrightarrow{1 \otimes \mu} R \otimes_{\mathbb{Z}} \mathcal{A}b(M, N) \xrightarrow{\mu} & \mathcal{A}b(M, N) \end{array}$$

Tensoring the top row by $- \otimes_{\mathbb{Z}} M$ and postcomposing by the counit yields the solid arrows

$$\begin{array}{ccc} (R \otimes_{\mathbb{Z}} R) \otimes_{\mathbb{Z}} \mathcal{A}b(M, N) \otimes_{\mathbb{Z}} M & \xrightarrow{m_R \otimes 1 \otimes 1} R \otimes_{\mathbb{Z}} \mathcal{A}b(M, N) \otimes_{\mathbb{Z}} M & \xrightarrow{1 \otimes \varepsilon} R \otimes_{\mathbb{Z}} N \\ & \searrow \lambda & \downarrow \rho_N \\ & & N \end{array}$$

The dotted arrows are explained by noting $\varepsilon = \text{ev}$ and applying the definition of λ . Doing the same for the bottom row and applying $\lambda = \varepsilon \circ (\mu \otimes 1)$ yields

$$\begin{array}{ccccccc} R \otimes_{\mathbb{Z}} (R \otimes_{\mathbb{Z}} \mathcal{A}b(M, N)) \otimes_{\mathbb{Z}} M & \xrightarrow{1 \otimes \mu \otimes 1} & R \otimes_{\mathbb{Z}} \mathcal{A}b(M, N) \otimes_{\mathbb{Z}} M & \xrightarrow{\mu \otimes 1} & \mathcal{A}b(M, N) \otimes_{\mathbb{Z}} M & \xrightarrow{\varepsilon} & N. \\ & & & & \searrow \lambda & \nearrow & \\ & & & & & & \end{array}$$

We can write λ using the definition, yielding a diagram

$$\begin{array}{c}
\begin{array}{ccccccc}
R \otimes_{\mathbb{Z}} (R \otimes_{\mathbb{Z}} \mathcal{A}b(M, N)) \otimes_{\mathbb{Z}} M & \xrightarrow{1 \otimes \mu \otimes 1} & R \otimes \mathcal{A}b(M, N) \otimes M & \xrightarrow{\mu \otimes 1} & \mathcal{A}b(M, N) \otimes M & \xrightarrow{\varepsilon} & N \\
\downarrow 1 \otimes 1 \otimes \varepsilon & \searrow 1 \otimes (\varepsilon \circ (\mu \otimes 1)) & \downarrow 1 \otimes \varepsilon & & & \nearrow \rho_N & \\
R \otimes R \otimes N & \xrightarrow{\rho_N} & R \otimes N & & & &
\end{array} \\
\lambda
\end{array}$$

The arrow $1 \otimes \lambda$ follows by functoriality of $R \otimes -$, and there are implicit composites with the associator.

3. The Proof: We will show that the transpose of the **Goal** diagram is the **Given** diagram (pasted with ev).

Transpose the Top Path of (1): We append $(- \otimes 1_M)$ and then ev .

$$(R \otimes R) \otimes H \otimes M \xrightarrow{(m_R \otimes 1) \otimes 1_M} (R \otimes H) \otimes M \xrightarrow{\mu \otimes 1_M} H \otimes M \xrightarrow{\text{ev}} N$$

By pasting diagram **(2)** (the adjunction identity), we replace the path $\text{ev} \circ (\mu \otimes 1_M)$ with λ :

$$(R \otimes R) \otimes H \otimes M \xrightarrow{(m_R \otimes 1) \otimes 1_M} (R \otimes H) \otimes M \xrightarrow{\lambda} N$$

Now, by pasting diagram **(1)** (your definition), we replace λ with $\rho_N \circ (1_R \otimes \text{ev})$:

$$(R \otimes R) \otimes H \otimes M \xrightarrow{(m_R \otimes 1) \otimes 1_M} (R \otimes H) \otimes M \xrightarrow{1_R \otimes \text{ev}} R \otimes N \xrightarrow{\rho_N} N$$

This simplifies (by naturality) to: $\rho_N \circ (m_R \otimes \text{ev})$. This is the transpose of the top path.

Transpose the Bottom Path of (1):

$$(R \otimes R) \otimes H \otimes M \xrightarrow{(1_R \otimes \mu) \otimes 1_M} (R \otimes H) \otimes M \xrightarrow{\mu \otimes 1_M} H \otimes M \xrightarrow{\text{ev}} N$$

Paste **(2)**: $\text{ev} \circ (\mu \otimes 1_M)$ becomes λ :

$$(R \otimes R) \otimes H \otimes M \xrightarrow{(1_R \otimes \mu) \otimes 1_M} (R \otimes H) \otimes M \xrightarrow{\lambda} N$$

Paste **(1)**: λ becomes $\rho_N \circ (1_R \otimes \text{ev})$:

$$(R \otimes R) \otimes H \otimes M \xrightarrow{(1_R \otimes \mu) \otimes 1_M} (R \otimes H) \otimes M \xrightarrow{1_R \otimes \text{ev}} R \otimes N \xrightarrow{\rho_N} N$$

This simplifies (by naturality) to: $\rho_N \circ (1_R \otimes (\text{ev} \circ (\mu \otimes 1_M)))$. Now paste **(2)** again inside the parentheses: $\text{ev} \circ (\mu \otimes 1_M)$ becomes λ :

$$(R \otimes R) \otimes H \otimes M \xrightarrow{\quad} R \otimes (H \otimes M) \xrightarrow{1_R \otimes \lambda} R \otimes N \xrightarrow{\rho_N} N$$

Finally, paste **(1)** again: λ becomes $\rho_N \circ (1_R \otimes \text{ev})$:

$$(R \otimes R) \otimes H \otimes M \xrightarrow{\quad} R \otimes (H \otimes M) \xrightarrow{1 \otimes (1 \otimes \text{ev})} R \otimes (R \otimes N) \xrightarrow{1_R \otimes \rho_N} R \otimes N \xrightarrow{\rho_N} N$$

This simplifies to $\rho_N \circ (1_R \otimes \rho_N) \circ (1_R \otimes 1_R \otimes \text{ev})$. This is the transpose of the bottom path.

****Conclusion:**** You have shown that the two paths in the transposed diagram are: $\rho_N \circ (m_R \otimes \text{ev})$
 $\rho_N \circ (1_R \otimes \rho_N) \circ (1_R \otimes 1_R \otimes \text{ev})$

The ****Given**** diagram (2) states that $\rho_N \circ (m_R \otimes 1_N) = \rho_N \circ (1_R \otimes \rho_N) \circ a_N$. If we pre-compose this known identity with $(1_R \otimes 1_R \otimes \text{ev})$, we get exactly that your two transposed paths are equal.

Since the transposed paths are equal, the original paths in the ****Goal**** diagram (1) must be equal by the properties of the adjunction. You can use this exact "pasting" method for the unitality axiom. ■

Problem 3

Problem 3.1. Let R be a ring, M an R -module, and $\pi : M \rightarrow M$ an R -linear map with $\pi^2 = \pi$. Show that $M \cong \text{Ker } \pi \oplus \text{Im } \pi$.

Proof. This is equivalent to saying that the exact sequence

$$0 \longrightarrow \text{Ker } \pi \longrightarrow M \longrightarrow \text{Im } \pi \longrightarrow 0$$

splits. By the splitting lemma, it suffices to show that π has a section. By hypothesis, the restriction $\pi|_{\text{Im } \pi} = 1$, and since π is an R -linear map, the inclusion $i : \text{Im } \pi \hookrightarrow M$ is R -linear. Therefore $\pi \circ i = 1$. ■

Problem 4

Problem 4.1. Let R a ring, $n > 1$. Include $R^{\oplus n-1} \hookrightarrow R^{\oplus n}$ via $(r_1, \dots, r_{n-1}) \mapsto (r_1, \dots, r_{n-1}, 0)$. Prove that

$$\frac{R^{\oplus n}}{R^{\oplus n-1}} \cong R.$$

Proof. The projection $\pi_n : R^{\oplus n} \rightarrow R$ onto the n th element of each tuple has $R^{\oplus n-1}$ as its kernel and is surjective. The statement follows by the first isomorphism theorem. ■

Problem 6

Problem 5.1. Let $\{M_\alpha\}_{\alpha \in A}$ and N be R -modules. If A is finite, show

1.

$$R\text{-Mod} \left(N, \bigoplus_{\alpha \in A} M_\alpha \right) \cong \bigoplus_{\alpha \in A} R\text{-Mod}(N, M_\alpha)$$

2.

$$R\text{-Mod} \left(\bigoplus_{\alpha \in A} M_\alpha, N \right) \cong \bigoplus_{\alpha \in A} R\text{-Mod}(M_\alpha, N).$$

Find counterexamples if A is infinite.

Proof. In any locally small category, the hom functor is continuous in both arguments; in the contravariant argument this means it sends colimits in $R\text{-Mod}$ to limits in Set . Since finite coproducts and finite products coincide in $R\text{-Mod}$, the claim is immediate. [na have to view as internal hom use tensor-hom adj and dual adj for (2)?]

For a counterexample to (2), we consider $\mathbb{R}$ -vector spaces. The arbitrary coproduct over a family $\{M_\alpha\}_{\alpha \in A}$ of $\mathbb{R}$ -vector spaces, which we will always denote $\oplus$, is composed of finite $\mathbb{R}$ -linear combinations in $\{M_\alpha\}_{\alpha \in A}$. The product is the set of all sequences $\{m_\alpha \in M_\alpha\}_{\alpha \in A}$, with operations defined pointwise.

Consider the product and coproduct

$$\mathbb{R}^{\mathbb{N}} := \prod_{n \in \mathbb{N}} \mathbb{R}, \quad \mathbb{R}^{\oplus \mathbb{N}} := \bigoplus_{n \in \mathbb{N}} \mathbb{R}.$$

Note that since $\mathbb{N}$ has cardinality $\aleph_0$, by definition of dimension the space $\mathbb{R}^{\oplus \mathbb{N}}$ has dimension $\aleph_0$. We show that $\mathbb{R}^{\mathbb{N}}$ has dimension $\mathfrak{c} = |\mathbb{R}|$. Since the cardinality of $\mathbb{R}^{\mathbb{N}}$ is $\aleph_0 \cdot \mathfrak{c} = \mathfrak{c}$, we have $\dim \mathbb{R}^{\mathbb{N}} \leq \mathfrak{c}$, so it suffices to construct a linearly independent set of cardinality $\mathfrak{c}$. Define $v_r = \{r^n\}_{n \in \mathbb{N}}$ for all $0 < r < 1$. The cardinality of the set $\{v_r\}_{0 < r < 1}$ is $\mathfrak{c}$, so we show the set is linearly independent. Consider a finite linear combination

$$\sum_{i=1}^n c_i v_{r_i} = 0$$

with distinct r_i s. This is equivalent to an infinite system of equations

$$\begin{cases} \sum_{i=1}^n c_i = 0 \\ \sum_{i=1}^n c_i r_i = 0 \\ \sum_{i=1}^n c_i r_i^2 = 0 \\ \vdots \end{cases}.$$

Since the linear combination is finite, we can represent the first n equations as the matrix equation

$$\begin{pmatrix} 1 & 1 & \cdots & 1 \\ r_1 & r_2 & \cdots & r_n \\ \vdots & \vdots & \ddots & \vdots \\ r_1^{n-1} & r_2^{n-1} & \cdots & r_n^{n-1} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = 0.$$

The relevance of this is that this is a *square matrix*, and in particular a Vandermonde matrix¹, which have determinant

$$\prod_{0 \leq i < j \leq n-1} (r_j - r_i).$$

Since the r_i s are distinct, the determinant is nonzero, so the matrix is invertible. This shows that all of the c_i s are 0. Therefore, the set is linearly independent, so $\dim \mathbb{R}^{\mathbb{N}} \geq \mathfrak{c}$.

¹The usual form, including the one given on the Wikipedia page, is the transpose of the given form. Determinants are invariant under transpose.

Since $\mathbb{N}$ is a set, there are natural isomorphisms

$$\begin{aligned}\mathbb{R}\text{-Vect}(\mathbb{R}^{\oplus \mathbb{N}}, \mathbb{R}) &= \mathbb{R}\text{-Vect}\left(\bigoplus_{n \in \mathbb{N}} \mathbb{R}, \mathbb{R}\right) \\ &\cong \prod_{n \in \mathbb{N}} \mathbb{R}\text{-Vect}(\mathbb{R}, \mathbb{R}) \\ &\cong \prod_{n \in \mathbb{N}} \mathbb{R} \\ &= \mathbb{R}^{\mathbb{N}}.\end{aligned}$$

The second isomorphism follows by noting that since $1 \cong - \otimes_{\mathbb{R}} \mathbb{R} \dashv \mathbb{R}\text{-Vect}(\mathbb{R}, -)$, by $1 \dashv 1$, and by uniqueness of adjunctions up to isomorphism, we have $\mathbb{R}\text{-Vect}(\mathbb{R}, -) \cong 1$. We then assemble the isomorphisms $\mathbb{R}\text{-Vect}(\mathbb{R}, \mathbb{R}) \cong \mathbb{R}$ along the $\mathbb{N}$ -ary product multifunctor. Since $\mathbb{R}^{\mathbb{N}}$ is not isomorphic to $\mathbb{R}^{\oplus \mathbb{N}}$, this gives us a counterexample for (2).

For a counterexample to (1), [i just redid 2 bruh] consider $\mathbb{Z}$ as a $\mathbb{Z}$ -module. The claim is that

$$\mathcal{A}b\left(\bigoplus_{n \in \mathbb{N}} \mathbb{Z}, \mathbb{Z}\right) \cong \bigoplus_{n \in \mathbb{N}} \mathcal{A}b(\mathbb{Z}, \mathbb{Z}).$$

Note that there are natural isomorphisms $\mathcal{A}b(\mathbb{Z}, \mathbb{Z})$ and

$$\mathcal{A}b\left(\mathbb{Z}^{\oplus \mathbb{N}}, \mathbb{Z}\right) \cong \prod_{n \in \mathbb{N}} \mathcal{A}b(\mathbb{Z}, \mathbb{Z}) \cong \prod_{n \in \mathbb{N}} \mathbb{Z}.$$

The statement now reduces to showing there is no isomorphism

$$\prod_{n \in \mathbb{N}} \mathbb{Z} \cong \bigoplus_{n \in \mathbb{N}} \mathbb{Z}.$$

For an isomorphism to exist, the sets must have the same cardinality. We know that

$$\left|\prod_{n \in \mathbb{N}} \mathbb{Z}\right| = \aleph_0^{\aleph_0} \geq 2^{\aleph_0} > \aleph_0.$$

We can write $\mathbb{Z}^{\oplus \mathbb{N}}$ as the union over the following sets:

$$\mathbb{Z}^{\oplus \mathbb{N}} = \bigcup_{k \in \mathbb{N}} \left\{ \{a_n\}_{n \in \mathbb{N}} \subseteq \mathbb{Z} \mid a_n = 0 \forall n \geq k \right\}.$$

This follows since all elements can be viewed as collections of elements where all but finitely many are zero. In particular, the set of $n \in \mathbb{N}$ such that a_n is nonzero is a *finite* subset of $\mathbb{N}$, and thus has a maximal element. The set $\{a_n\}_{n \in \mathbb{N}}$ then belongs to the union by taking $k = n + 1$. We can compute the cardinality of this by noting that each of the sets we are unioning over has cardinality $\aleph_0$, so

$$\left|\mathbb{Z}^{\oplus \mathbb{N}}\right| = \aleph_0^2 = \aleph_0.$$

Therefore, the groups are not even isomorphic as sets, and thus cannot be isomorphic as $\mathbb{Z}$ -modules. ■